

Database Management Systems - IT1223

Mr.M.N.M Shakoor, DPS,FAS, University of Vavuniya

Retrieving Information from a Table

- The `SELECT` statement is used to pull information from a table. The general form of the statement is:

- **Syntax:**

```
SELECT what_to_select
FROM which_table
WHERE conditions_to_satisfy;
```

1. Selecting All Data:

- The simplest form of `SELECT` retrieves everything from a table:
 - `SELECT * FROM pet;`

2. Selecting Particular Rows:

- Write a query to select Bowser's record:
 - `SELECT * FROM pet WHERE name = 'Bowser';`
 - Write a query to know which animals were born during or after 1998:
 - `SELECT * FROM pet WHERE birth >= '1998-1-1';`
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- The MySQL AND, OR and NOT Operators

- Write a query to locate female dogs:

- `SELECT * FROM pet WHERE species = 'dog' AND sex = 'f';`

- Write a query to locate species of snake or bird:

- `SELECT * FROM pet WHERE species = 'snake' OR species = 'bird';`

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- `AND` , `OR` can be intermixed, although `AND` has higher precedence than `OR`. If you use both operators, it is a good idea to use parentheses to indicate explicitly how conditions should be grouped:

- `SELECT * FROM pet WHERE (species = 'cat' AND sex = 'm') OR (species = 'dog' AND sex = 'f');`

3. Selecting Particular Columns

- Write a query to know when your animals were born:

- `SELECT name, birth FROM pet;`

- Write a query to know all the owners of the pets:

- `SELECT owner FROM pet;`

➤ MySQL `SELECT DISTINCT` Statement

- The `SELECT DISTINCT` statement is used to return only distinct (different) values.

- `SELECT DISTINCT owner FROM pet;`

➤ Combine row selection with column selection using `WHERE`.

- Write a query to get birth dates for dogs only:

- `SELECT name, species, birth FROM pet WHERE species = 'dog' OR species = 'cat';`

4. Sorting Rows

- Write a query to display animal and birthdays according to the birthday's order (Ascending and Descending)
 - `SELECT name, birth FROM pet ORDER BY birth;`
 - `SELECT name, birth FROM pet ORDER BY birth DESC;`
 - You can sort on multiple columns, and you can sort different columns in different directions. For example, to sort by type of animal in ascending order, then by birth date within animal type in descending order (youngest animals first), use the following query:
 - `SELECT name, species, birth FROM pet ORDER BY species, birth DESC;`
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5. Date Calculations

- Write a query to determine how many years old of each pets (use the `TIMESTAMPDIFF()` function)
 - `SELECT TIMESTAMPDIFF(MONTH, '2003-02-01', '2003-05-01');` // Different in Months
 - `SELECT TIMESTAMPDIFF(YEAR, '2002-05-01', '2001-01-01');` // Different in years
 - `SELECT TIMESTAMPDIFF(MINUTE, '2003-02-01', '2003-05-01 12:05:55');` // Different in Minutes

Query: `SELECT name, birth, CURDATE(),
TIMESTAMPDIFF(YEAR, birth, CURDATE()) AS age
FROM pet;`

- Write a query to display pets name, birth day and age according to the pet's name order (Ascending)
- Write a query to display pets name, birth day and age according to the pet's age order (Ascending)

```
SELECT name, birth, CURDATE(), TIMESTAMPDIFF(YEAR, birth, CURDATE()) AS age FROM pet ORDER BY name;
```

```
SELECT name, birth, CURDATE(), TIMESTAMPDIFF(YEAR, birth, CURDATE()) AS age FROM pet ORDER BY age;
```

- Write a query to determine age at death for animals that have died.

```
SELECT name, birth, death,  
TIMESTAMPDIFF(YEAR, birth, death) AS age  
FROM pet WHERE death IS NOT NULL ORDER BY age;
```

- What if you want to know which animals have birthdays next month?

- First understand these queries:

- `SELECT name, birth, MONTH(birth) FROM pet;`
- `SELECT name, birth FROM pet WHERE MONTH(birth) = 5;`

➤ But what if the current month is December. You cannot add one to the month 12 and look for animals born in month 13, because there is no such month.

- `SELECT name, birth FROM pet
WHERE MONTH(birth) = MONTH (DATE_ADD (CURDATE () , INTERVAL 1 MONTH)) ;`

Or try this

- `SELECT name, birth FROM pet
WHERE MONTH(birth) = MOD (MONTH (CURDATE ()) , 12) + 1;`
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♠ *If a calculation uses invalid dates, the calculation fails and produces warnings:*

- `SELECT '2018-10-31' + INTERVAL 1 DAY;`
 - `SELECT '2018-10-32' + INTERVAL 1 DAY;`
 - `SHOW WARNINGS;`
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To be Continued..!
